GREEN CITY {HACKATHON_}

{ DATA FOR THE HEILBRONN OF TOMORROW. }

> WHEN: **TUESDAY 27 – THURSDAY 29 OCTOBER 2026**

> WHERE: **OPEN SPACE HEILBRONN**

/ PHOTOGRAPHY AT THIS EVENT

NOTICE

Please be aware that photographs will be taken during this event.

If you do not wish to be photographed, please speak to a member of our staff.

> **Thank you for your understanding.**

OpenSpace Heilbronn · Arkadia Heilbronn gGmbH

Schedule

What we are doing the next couple of days

/ SCHEDULE

GREEN CITY HACKATHON

TUE · 27 OCT 2026

Kick-off & orientation

Registration, welcome and an introduction to the data and sensors. Pitch your ideas, form teams and start hacking.

WED · 28 OCT 2026

Work

A full day for coding, building and prototyping your ideas

THU · 29 OCT 2026

Wrap-up & pitch

Last minute efforts, selection of the finalists and presentation in front of the jury

/ AGENDA

GREEN CITY {HACKATHON_}

01 INTRO - GREEN CITY HEILBRONN

02 GREEN CAPITAL HEILBRONN

03 CITIZEN SCIENCE

04 HARDWARE & DATA

05 CHALLENGES & GOALS

06 COMMUNICATION

/ MENTORS - re:edu

Ask us anything !

Umut Tas

Impact

Jan Wirwahn

Hardware

Eric Thieme-Garmann

Software

Felix Erdmann

Data

Dr. Steffen Ciprina

Education

Eva Jacobs

Education

/ RE:EDU - reengineering education

Where we come from

01 DIGITAL EDUCATION

> Digital education with senseBox
> Learning concepts and workshops

02 CITIZEN SCIENCE

> Technologies for citizen science
> Workshops and campaigns
> Data-driven civic participation

03 SMART CITIES

> Open data dashboards and visualizations
> Measurement Networks

/ MENTORS - 42 HEILBRONN

Ask us anything !

Jonas Götz

42 Heilbronn

Rayan Mehadjeri

42 Heilbronn

Maximilian Kneisl

42 Heilbronn

/ ORGA TEAM

Marianna Georgiou

Open Space Managerin

Alexander Gantikow

AIM

/ 01

Green City Heilbronn

Our contribution to a greener Heilbronn.

/ GREEN CITY HEILBRONN

GREEN CITY HACKATHON

re:edu reengineering education

aim INNOVATIVE BILDUNG

ARKADIA

A **collaborative Project** to bring innovative education to Heilbronn.

With:

- real **Data**
- real **Problems**
- real **People**
- (and hopefully) real **Solutions**

/ Campaigns

Data gathered by people living in Heilbronn

HEAT ISLANDS

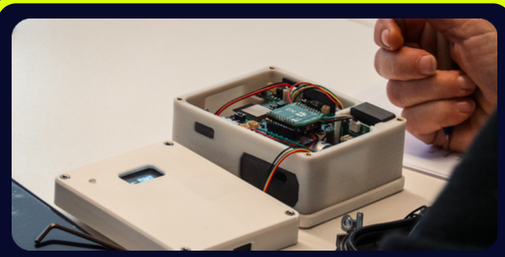
August 2026

> 30 students

BIRDHOUSE-WORKSHOP

JUNI 2026

> 15 citizens



BIKE-WORKSHOPS

JUNI 2026

> 25 senseBox:bike’s

CHECK-DIE-LUFT (AQ)

MÄRZ 2026

> Fine dust map

AIR QUALITY WEEK

MÄRZ 2026

> 16 Schools

HACKYOURCITY

HERBST 2025

> Autumnschool

SENSEBOX:BIKE

SEIT DEZ. 2025

> 12 senseBox:bike’s

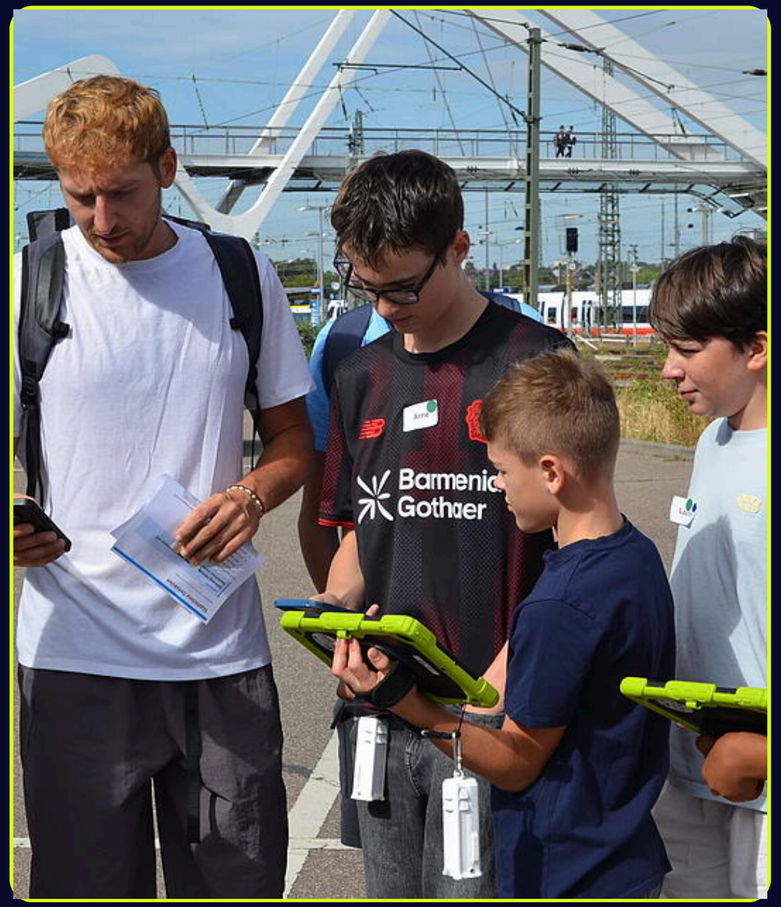
NEXT UP?

greencity.hn ↗

/ SUMMERSCHOOL HEATISLANDS 2026

Young researchers map Heilbronn’s heat islands

GREEN CITY {HACKATHON_}



73 km
DISTANCE COVERED

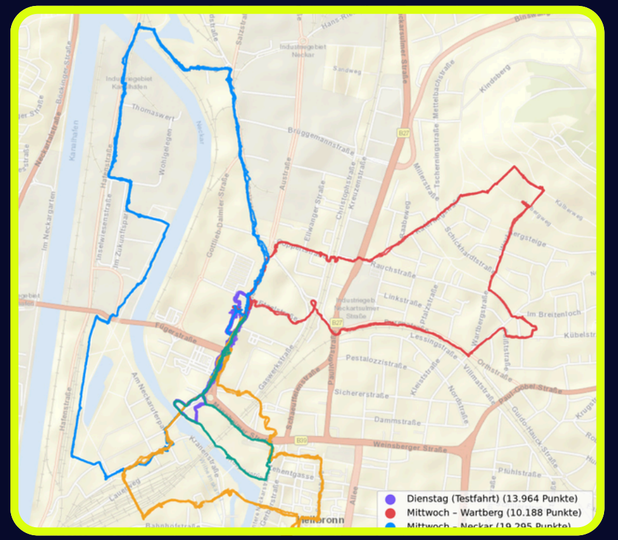
61.546
DATA POINTS

9
DEVICES

About

At the Heat Island Summer School, young people aged 11 to 16 measured temperatures across Heilbronn with the senseBox and created their own maps.

HUMIDITY TEMPERATURE



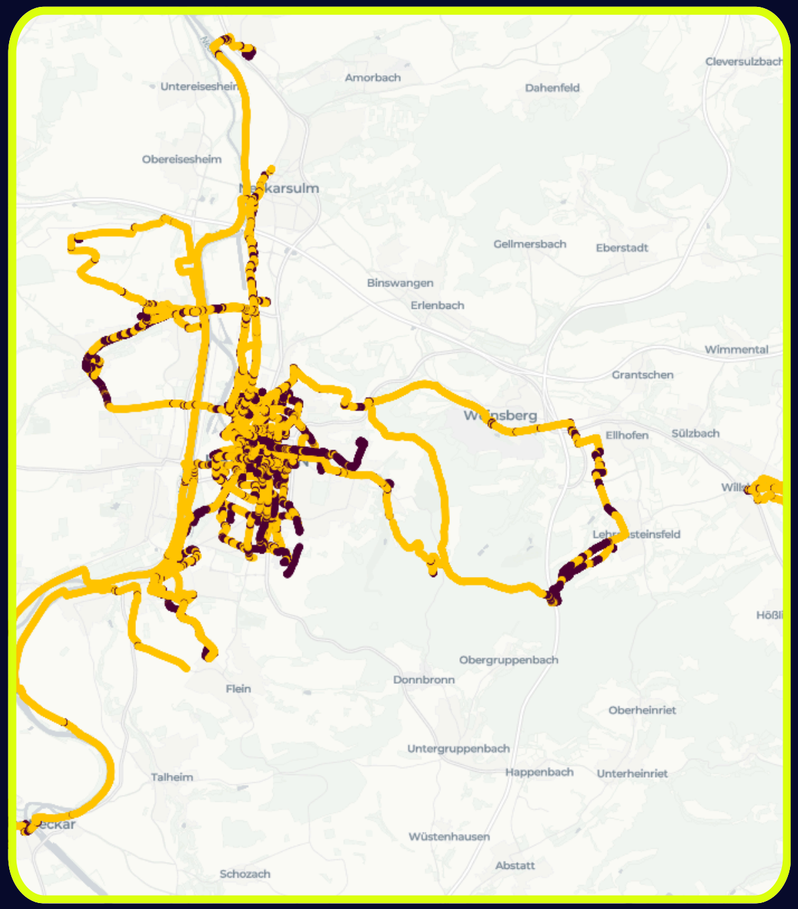
/ SENSEBOX:BIKE

Seit Dezember 2025 auf Heilbronns Radwegen unterwegs.

About

The sensebox:bike is an enviromental measurement station which is perfected for the usage on (any) bike. More information on **https://sensebox.de/en/products-bike**

FINE DUST | TEMPERATURE | OVERTAKING DISTANCE



/ SENSEBOX:BIKE IN NUMBERS

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Data collected on Heilbronn's bike lanes since December 2025.

5.200 km
TOTAL DISTANCE

2.263
TOTAL RIDES

31
RIDERS

303 h
TOTAL TIME

17 km/h
AVG. SPEED

2.3 km
AVG. RIDE DISTANCE

8 min
AVG. RIDE TIME

64 km
LONGEST RIDE

{ 5.200 km ≈ 13 % of the way around the Earth }

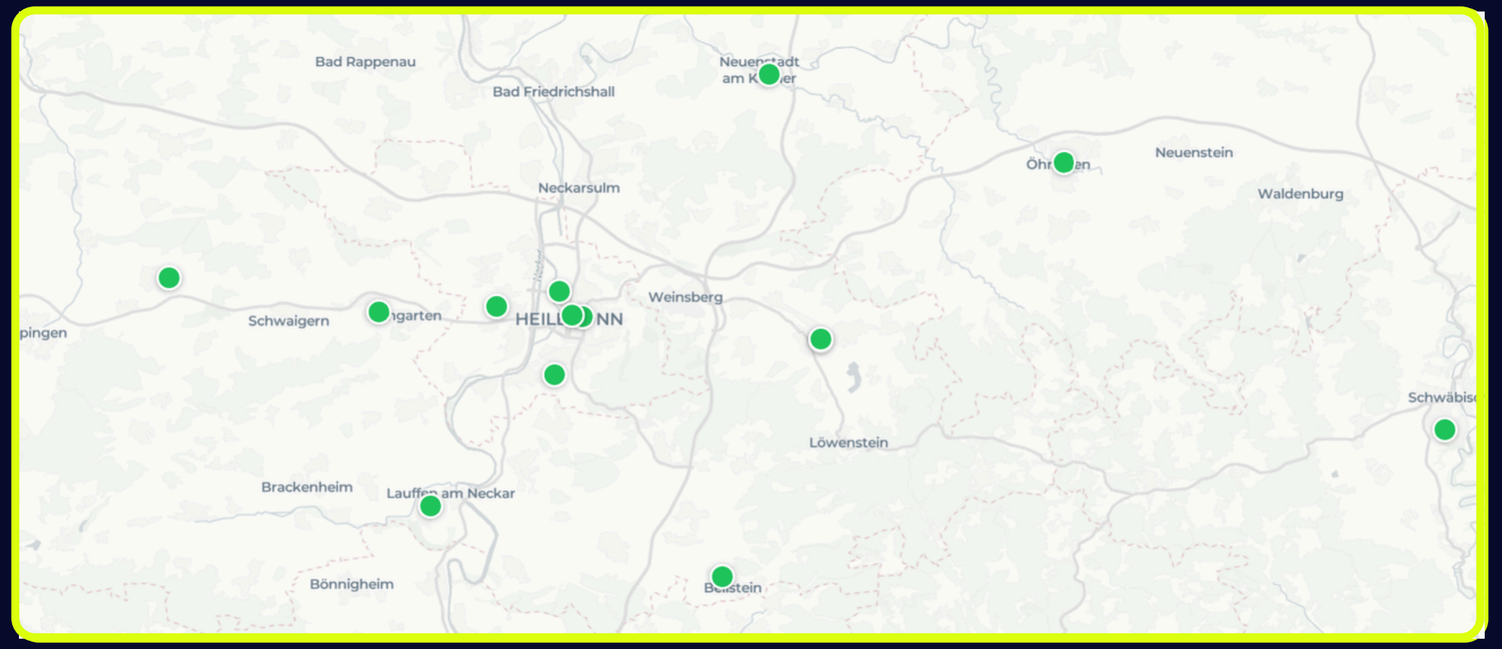
/ SENSEBOX:HOME

Schools across Heilbronn keep an eye on the air.

15 SCHOOLS WITH A SENSEBOX:HOME

24/7 MEASURING AROUND THE CLOCK

PM FINE DUST MONITORING

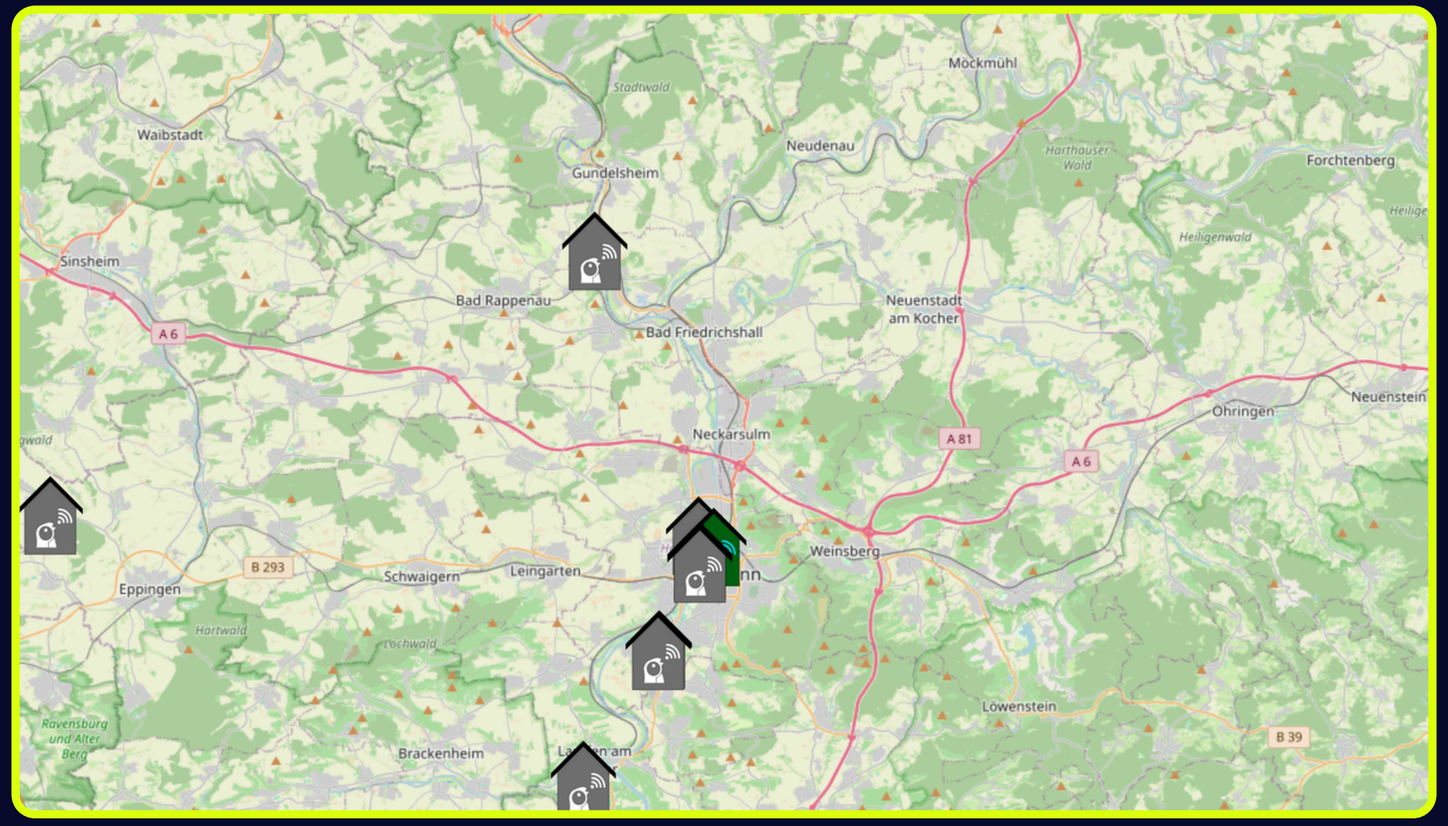


Each school runs its own station and measures fine dust at a fixed spot. Together they show how good the air is in different parts of the city.

/ SMART BIRDHOUSES

GREEN CITY HACKATHON

Bird feeding station as a monitoring device for your garden



15 NEW BIRD RESEARCHERS IN HEILBRONN

The participants assembled the stations themselves and gathered data about birds living in or around their backyard

> All stations on the map: wiediversistmeingarten.org/view ↗

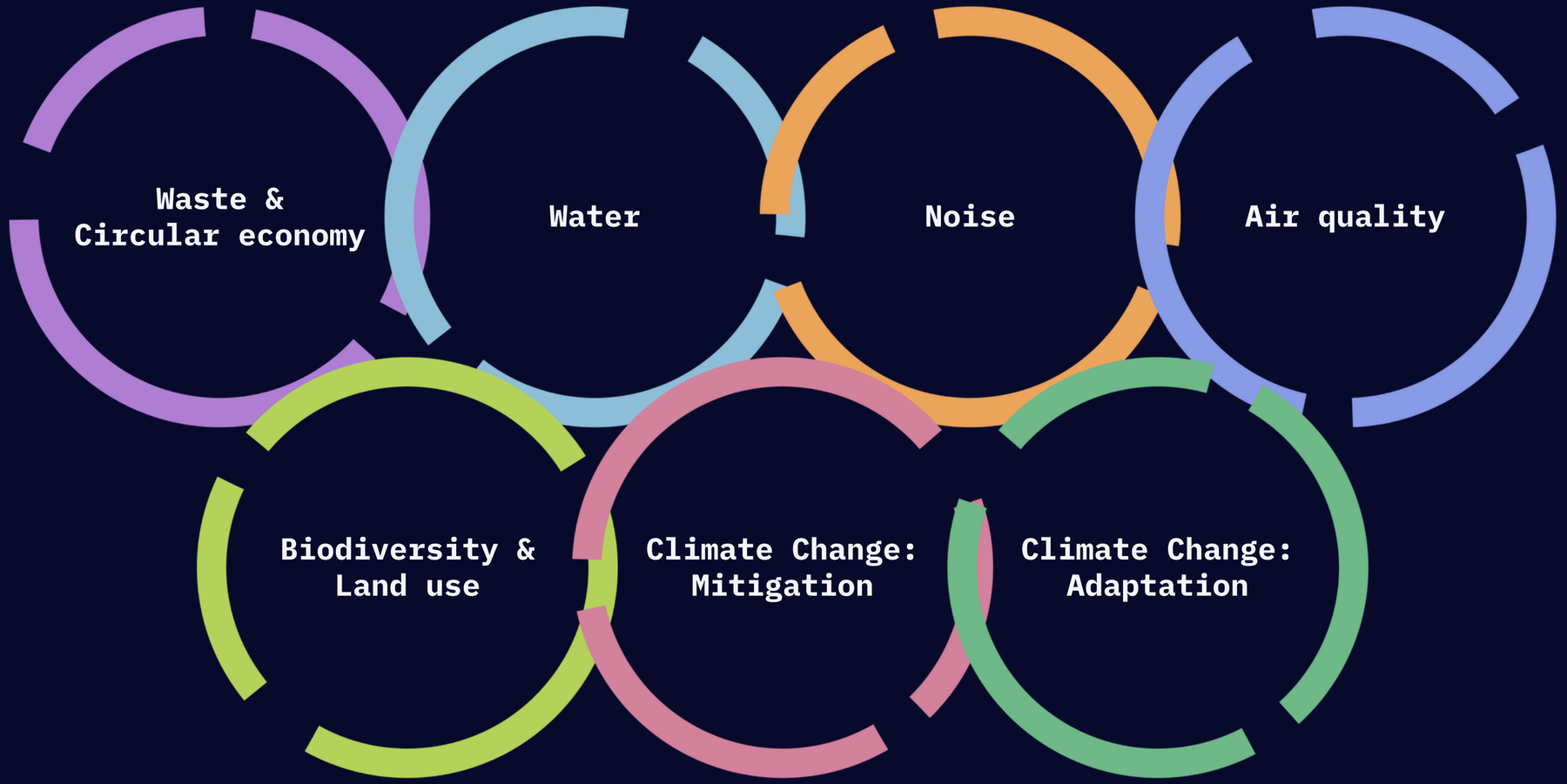
/ 02

Green Capital Heilbronn

Heilbronn becomes European Green Capital 2027.
Help to shape the city!

/ GREEN CAPITAL HEILBRONN

Topics of Green Capital Heilbronn 2027



/ HEILBRONN

European Green Capital 2027

GREEN CITY HACKATHON

01

TRANSFORMATION

From industrial site to green living space: new bike lanes, urban gardens, innovative neighborhoods and much more

02

PARTICIPATION

Citizens actively participate and help foster projects which tackle real world problems they encounter

03

2026

Building the foundation for a green year 2027

/ 03
Citizen Science

Only if we **understand**, can we care. Only if we **care**, we will help. Only if we **help**, we shall be **saved**

> Jane Goodall - World renowned ethologist and activist

/ HOW CAN DATA HELP?

From opinions to evidence

GREEN CITY HACKATHON

01

REAL PEOPLE, REAL PROBLEMS

People measure phenomena **where they live, cycle or walk**. So the data shows what they really go through: how hot their street gets, the air on the way to school, how close cars pass on the bike lane. That makes problems visible and hard to ignore.

e.g. **During the summer we were able to identify heat islands and tell the municipality where to plant more trees**

02

QUANTITY OVER PERFECTION

Low-cost sensors are not lab equipment. But lots of measurements from lots of people still **show patterns like hotspots, trends and outliers**. Therefore we can spot problems more locally and come up with ideas to fix them.

e.g. **With 2.623 rides with the senseBox:bike we have a vast amount of mobility data.**

/ 04
Hardware & Data

/ SENSEBOX:BIKE

Bike-mounted sensor box that logs environmental data, street surface type and overtaking cars while riding through the city.

MEASURES

AIR QUALITY (PM) · TEMPERATURE · HUMIDITY · STREET SURFACE TYPE · OVERTAKING LIKELIHOOD · OVERTAKING DISTANCE · GPS TRACKS

DATA ENDPOINTS

> Data guide ↗
> Map with data ↗

DOCUMENTATION

> About the device ↗

/ MOBILE DATALOGGER

Mobile temperature and humidity sensor, to be carried around while walking through the city.

MEASURES

TEMPERATURE HUMIDITY

DATA ENDPOINTS

> Heat-Island-Summerschool 2026 ↗

Logged walks from the Heat-Island Summerschool 2026.

DOCUMENTATION

> About the device ↗

/ SENSEBOX:HOME

Stationary weather stations run by residents and schools, measuring around the clock.

MEASURES

TEMPERATURE · HUMIDITY · AIR PRESSURE · AIR QUALITY (PM)

DATA ENDPOINTS

> senseBox:home data ↗

Data of all registered senseBoxes in Heilbronn.

DOCUMENTATION

> Hardware documentation ↗

/ SMART BIRDHOUSES

Smart bird houses all over Heilbronn that log visits and conditions, giving a window into how urban wildlife responds to the changing climate.

MEASURES

BIRD PICTURES · VISITS · TEMPERATURE · HUMIDITY

DATA ENDPOINTS

> Starter notebook (Colab) ↗

Python notebook that gives a brief overview of the Birdiary API.

> Starter notebook (GitHub) ↗

Alternative link to the same notebook.

DOCUMENTATION

> Project page ↗
> Map of all stations ↗
> Code ↗
> Green City Dashboard ↗

/ SMART BEEHIVE

Sensor-equipped beehive at the OpenSpace, plus experimental entrance videos with labelled bee activity for machine learning.

MEASURES

TEMPERATURE · HUMIDITY · ENTRANCE VIDEOS · BEE ACTIVITY LABELS

DATA ENDPOINTS

> Sensor data ↗

Environmental data inside and outside of three beehives around the OpenSpace.

> Labelled entrance videos ↗

Beehive entrance videos recorded with the senseBox MCU Eye. Potentially useful for machine learning.

/ 05

Challenges & Goals

Ein Satz, der beschreibt, worum es in diesem Abschnitt geht.

/ THE CHALLENGES

What you'll be working on for the next three days.

GREEN CITY HACKATHON

01

START WITH A QUESTION

Something about Heilbronn you've noticed, wondered about or been annoyed by.

e.g. **Which streets get the hottest in summer?**

02

ANSWER IT WITH DATA

Use the data that's already there, or go out and collect your own. Both count.

e.g. **Bike rides, birdhouses, open data, your own sensors**

03

MAKE IT USEFUL

Your result should help the people who live here. And that includes you.

e.g. **A map, an app or a website**

{ ALL ABOUT GREEN CAPITAL HEILBRONN }

- Waste & circularity
- Water
- Noise
- Air quality
- Biodiversity & land use
- Climate mitigation
- Climate adaptation

/ YOUR GOAL

GREEN CITY {HACKATHON_}

Use the data to spot or tackle a real problem in Heilbronn.

01

PICK A TOPIC

- Waste & circular economy
- Water
- Noise
- Air quality
- Biodiversity & land use
- Climate change: mitigation
- Climate change: adaptation

+

02

USE THE DATA OR SENSORS

Analyse, visualise, combine or extend or collect your own data.

> Bike, home & datalogger
> Birdhouses & beehive
> Open data
> Your own measurements

→

03

BUILD SOMETHING

Any format works.

APPS · TOOLS · MAPS · WEBSITES · INFOGRAPHICS · DASHBOARD TILES · DATA ANALYSIS · MEASUREMENT DEVICES

{ Anything goes as long as it helps the city. Different idea? Ask a mentor! }

/ Judging Criteria

What will the judges be looking for?

01 Teamwork

02 Innovation & Originality

03 Functionality & Technical Execution

04 Impact and Feasability

05 User Experience & Design

06 Presentation & Demo

/ PRIZES

For the top 3 teams.

UP TO €250 IN AI CREDITS

OpenRouter access: ChatGPT, Claude, Gemini & co. Credits go into your project, so you can keep developing it.

€250

PITCH IT TO CITY HALL

Your own session with the Deputy Mayor. Real feedback from someone who helps shape what gets built in this city.

Date

EXCLUSIVE FOLLOW-UP COURSE

Sensors, hardware, software — whatever your project needs. Experts help you turn your idea into a product.

Course

/ 06
Communication

/ AWARENESS TEAM

Something doesn't feel right? Talk to us.

We want everyone to feel safe and welcome here.

If you feel uncomfortable, see something that isn't okay or just need someone to talk to, come find us. Or send us a message on Mattermost.

Eva

Awareness team

Steffen

Awareness team

/ BUILD YOUR TEAM

Find your people – and your idea.

GREEN CITY HACKATHON

4

MIN. MEMBERS

6

MAX. MEMBERS

> **Need more people for your idea?** Talk to the mentors. We'll figure something out

01 **FIND A TOPIC**

Pick a topic and an idea you want to work on & **talk to a Mentor to confirm the Idea.**

02 **CREATE YOUR TEAM**

Register it on the website and describe your idea.

03 **STILL LOOKING?**

Browse the teams, see what others are building – and ask to join.

hackathon.greencity.hn/teams

/ COMMUNICATION

Where to find information – and help.

01

MATTERMOST

Our chat for the hackathon. Reach all mentors and coordinate with your team.

> chat.greencity.hn ↗

02

WEBSITE

Schedule, topics, resources and all teams in one place.

> hackathon.greencity.hn ↗

03

ON SITE

Mentors are around the whole time – just ask them anything.

{ Everyone please join Mattermost! }

/ JOIN MATTERMOST

Scan the code and sign up – takes one minute.

chat.greencity.hn

01 **SCAN THE QR CODE**

Or open the link on your laptop.

02 **CREATE YOUR ACCOUNT**

Sign up with your name and email address.

03 **OPEN MATTERMOST**

Use it in your browser – or in the Mattermost app with the server chat.greencity.hn.

04 **SAY HI!**

Ask the mentors anything and coordinate with your team.

/ TODAY'S SCHEDULE

TUE · 27 OCT 2026

12:00 — **Lunch & Team formation**
Pizza

13:00 — **Pitches & Team formation**
Teams that still need members pitch their idea on stage (60 seconds). Teams form and agree on their plans with the mentors.

14:00 — **Activation training**

15:00 — **Work Phase 1**
Explore the data, ask your first questions and define your approach. Then get to work! Mentors are on hand.

15:00 — **Q&A Session**
Vote on Mattermost!!!

18:00 — **Dinner**
Catering

18:30 — **Work Phase**
Open End

/ GET CONNECTED

Two things to do right now. Grab your phone.

GREEN CITY {HACKATHON_}

01

JOIN MATTERMOST

Our chat for the hackathon. Talk to the mentors and your team.

> chat.greencity.hn ↗

02

REGISTER YOUR TEAM; TALK TO MENTORS FIRST

Create your team, or see what the others are building and join one.

> hackathon.greencity.hn/teams ↗

THANK YOU_

{ DATA FOR THE HEILBRONN OF TOMORROW. }

HACKATHON.GREENCITY.HN

ARKADIA

aim INNOVATIVE BILDUNG

re:edu reengineering education

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AKKREDITIERUNG